

Leica Viva TS16/60 Leica Nova MS60 **GeoCOM Reference Manual**

Version 1.50
English



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1 CHANGES FOR VIVA TS16 / NOVA TS60 / NOVA TM60 / NOVA MS60

1.1 INTRODUCTION

This document describes the changes in GeoCOM related to former Viva / Nova totalstations and multistations. There are only a few differences compared to the existing commands. **The changes described are applied to firmware versions equal and greater Captivate v. 5.00.** The Nova TM60 only supports these features with a firmware v.6.00 and higher.

Overview about the changes

Change	Description	affected messages
BAP_CONT_REF_STANDARD	Measurement program removed	17019
BAP_SetATRSetting	Command removed	17035
New FrameRates for video stream	added 20, 22 and 30 Hz	23675
New instrument family	Added new family types	5035

2 BASIC APPLICATIONS – BAP

2.1 CHANGED CONSTANTS AND TYPES

Distance measurement programs

```
enum BAP_USER_MEASPRG {
    BAP_SINGLE_REF_STANDARD = 0,    // IR Standard
    BAP_SINGLE_REF_FAST = 1,        // IR Fast
    BAP_SINGLE_REF_VISIBLE = 2,     // LO Standard
    BAP_SINGLE_RLESS_VISIBLE = 3,   // RL Standard
    BAP_CONT_REF_STANDARD = 4,      // IR Tracking
    BAP_CONT_REF_FAST = 5,          // not supported by Viva / Nova TPS
    BAP_CONT_RLESS_VISIBLE = 6,     // RL Fast Tracking
    BAP_AVG_REF_STANDARD = 7,       // IR Average
    BAP_AVG_REF_VISIBLE = 8,        // LO Average
    BAP_AVG_RLESS_VISIBLE = 9,      // RL Average
    BAP_CONT_REF_SYNCHRO = 10,      // IR Synchro Tracking
    BAP_SINGLE_REF_PRECISE = 11     // IR Precise (TS30, TM30)
};
```

BAP_CONT_REF_STANDARD was removed

Since Release 5.0 this enum value points by default to BAP_CONT_REF_SYNCHRO

2.2 CHANGED FUNCTIONS

2.2.1 BAP_SetMeasPrg - setting the distance measurement program

ASCII-Request

```
%R1Q,17019:4
```

ASCII-Response

```
%R1P,0,0:2
```

Remarks

If BAP_CONT_REF_STANDARD = 4 will be set then the response will be 2 (measurement program not available) – measurement program will not be changed, for IR Tracking the BAP_CONT_REF_SYNCHRO = 10 measurement program should be used.

Since Release 5.0 BAP_CONT_REF_STANDARD enum value points by default to BAP_CONT_REF_SYNCHRO

Parameters

eMeasPrg	In	Measurement program
----------	----	---------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
GRC_IVPARAM	2	Measurement program is not available

BAP_SetATRSetting – was removed

2.2.2 BAP_GetATRSetting – getting the current ATR low vis mode - removed

ASCII-Request

```
%R1Q,17034:
```

ASCII-Response

%R1Q,0,0:0,0

Remarks

Always returns BAP_LOWVIS_NORMAL (0)

Parameters

reATRSetting	Out	BAP_LOWVIS_NORMAL: ATR is using no special flags/modes BAP_LOWVIS_ON+ ATR low vis mode on BAP_LOWVIS_ALWAYS ON+ ATR low vis mode always on BAP_LOWVIS_BOBBY ON+ ATR high reflectivity mode on BAP_LOWVIS_BOBBYALWAYS ON+ ATR high reflectivity mode always on
--------------	-----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

2.2.3 BAP_SetATRSetting – setting the current ATR low vis mode**ASCII-Request**

%R1Q,17035: eATRSetting[long]

ASCII-Response

%R1Q,0,0:3081

Remarks

Always returns GRC_COM_PROC_UNAVAIL (3081)

Parameters

eATRSetting	In	ATR low vis mode
-------------	----	------------------

Return-Code Names and Return-Code Values

GRC_COM_PROC_UNAVAIL	3081	Unknown RPC, procedure ID invalid.
----------------------	------	------------------------------------

3 CAMERA – CAM

3.1 CHANGED FUNCTIONS

3.1.1 CAM_StartRemoteVideo – Start a remote video

ASCII-Request

%R1Q,23675:eCamID,nFrameRate,nQuality

ASCII-Response

%R1P,0,0:RC

Remarks

Parameter 2 added 20, 22 and 30 Hz

Parameters

eCamID	In	CAM_ID_TYPE
nFrameRate	In	[Hz] 3, 5, 10, 20, 22 and 30 allowed
nQuality	In	[kbps] Range between 100 kbps and 6144 kbps allowed

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

4 CENTRAL SERVICES – CSV

4.1 CHANGED FUNCTIONS

4.1.1 CSV_GetDeviceConfig – Get the instrument configuration

ASCII-Request

%R1Q, 5035:

ASCII-Response

%R1P, 0, 0:RC, *DevicePrecisionClass[long],*
 DeviceConfigurationType[long]

Remarks

New DeviceConfigurationType

Parameters

Device	Out	System information (see data type description for further information).
--------	-----	---

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

5 INSTRUMENT FAMILY – TPS

5.1 CHANGED CONSTANTS AND TYPES

```
// TPS Device Precision Class
enum TPS_DEVICE_CLASS {

    TPS_CLASS_1100 = 0,          // TPS1000 family member,
                                // 1 mgon, 3"
    TPS_CLASS_1700 = 1,          // TPS1000 family member,
                                // 0.5 mgon, 1.5"
    TPS_CLASS_1800 = 2,          // TPS1000 family member,
                                // 0.3 mgon, 1"
    TPS_CLASS_5000 = 3,          // TPS2000 family member
    TPS_CLASS_6000 = 4,          // TPS2000 family member
    TPS_CLASS_1500 = 5,          // TPS1000 family member
    TPS_CLASS_2003 = 6,          // TPS2000 family member
    TPS_CLASS_5005 = 7,          // TPS5000 family member
    TPS_CLASS_5100 = 8,          // TPS5000 family member

    TPS_CLASS_1102 = 100,        // TPS1100 family member, 2"
    TPS_CLASS_1103 = 101,        // TPS1100 family member, 3"
    TPS_CLASS_1105 = 102,        // TPS1100 family member, 5"
    TPS_CLASS_1101 = 103,        // TPS1100 family member, 1"

    TPS_CLASS_1202 = 200,        // TPS1200 family member, 2"
    TPS_CLASS_1203 = 201,        // TPS1200 family member, 3"
    TPS_CLASS_1205 = 202,        // TPS1200 family member, 5"
    TPS_CLASS_1201 = 203,        // TPS1200 family member, 1"

    TPS_CLASS_Tx30 = 300,        // TS30,TM30 family member, 0.5"
    TPS_CLASS_Tx31 = 301,        // TS30,TM30 family member, 1"

    TPS_CLASS_TDRA = 350,        // TDRA family member, 0.5"

    TPS_CLASS_TS01 = 500,        // Mid Range family member, 1"
    TPS_CLASS_TS02 = 501,        // Mid Range family member, 2"
    TPS_CLASS_TS03 = 502,        // Mid Range family member, 3"
    TPS_CLASS_TS05 = 503,        // Mid Range family member, 5"
    TPS_CLASS_TS06 = 504,        // Mid Range family member, 6"
    TPS_CLASS_TS07 = 505,        // Mid Range family member, 7"
    TPS_CLASS_TS10 = 506,        // Mid Range family member, 10"

    TPS_CLASS_TS1X_1 = 600,      // Viva TPS Family member, 1"
    TPS_CLASS_TS1X_2 = 601,      // Viva TPS Family member, 2"
    TPS_CLASS_TS1X_3 = 602,      // Viva TPS Family member, 3"
    TPS_CLASS_TS1X_4 = 603,      // Viva TPS Family member, 4"
    TPS_CLASS_TS1X_5 = 604,      // Viva TPS Family member, 5"

    TPS_CLASS_TX50_0_5 = 650,    // Nova Family member TS50/TM50, 0.5"
    TPS_CLASS_TX50_1 = 651,      // Nova Family member TS50/TM50,MS50 1"

    TPS_CLASS_TS16 = 900,        // Viva Family member TS16
    TPS_CLASS_TX60 = 910,        // Nova Family member TS60/MS60
};
```


6 CHANGES IN CAPTIVATE RELEASE 5.00

6.1 CHANGED CONSTANTS AND TYPES

6.1.1 CSV

```
// Hardware information
enum CSV_HW_GROUP {

    CSV_HW_GROUP_INVALID                = 0, // Invalid hardware group
    CSV_HW_GROUP_INTERNAL_RADIO_MODULE = 1, // Internal Radio
    CSV_HW_GROUP_INTERNAL_PHONE        = 2, // Internal Phone
    CSV_HW_GROUP_ANTENNA_ELEMENT       = 3, // Antenna Element for GS15
    CSV_HW_GROUP_TPS_ANGLE_ACCURACY    = 4, // TPS - Angle Accuracy
    CSV_HW_GROUP_TPS_EDM_TYPE          = 5, // TPS - EDM Type
    CSV_HW_GROUP_TPS_EGL               = 6, // TPS - EGL
    CSV_HW_GROUP_TPS_ATR               = 7, // TPS - ATR
    CSV_HW_GROUP_TPS_POWER_SEARCH      = 8, // TPS - Power Search
    CSV_HW_GROUP_TPS_OVERVIEW_CAMERA   = 9, // TPS - Overview Camera
    CSV_HW_GROUP_TPS_MOTORIZED         = 10, // TPS - Motorized
    CSV_HW_GROUP_TPS_DIODE_LASER       = 11, // TPS - DiodeLaser
    CSV_HW_GROUP_TPS_HOTSHOE           = 12, // TPS - HotShoe
    CSV_HW_GROUP_TPS_1ST_KEYBOARD      = 13, // TPS - 1st Keyboard
    CSV_HW_GROUP_TPS_2ND_KEYBOARD      = 14, // TPS - 2nd Keyboard
    CSV_HW_GROUP_TPS_PRISM_LOCK        = 15, // TPS - Prism Lock
    CSV_HW_GROUP_TPS_RL_MAX_DIST_LIM   = 16, // TPS - EDM RL Maximum
                                     // Distance Limitation
    CSV_HW_GROUP_TPS_ARCTIC            = 17, // TPS - Arctic Option
    CSV_HW_GROUP_TPS_ONAXIS_CAMERA     = 18, // TPS - On Axis Camera
    CSV_HW_GROUP_TPS_FOCUS_MODE        = 19, // TPS - Focus Mode
    CSV_HW_GROUP_TPS_WLAN              = 20, // TPS - WLAN
    CSV_HW_GROUP_TPS_SCANNING          = 21, // TPS - Scanning
    CSV_HW_GROUP_INTERNAL_ME           = 22, // CS - Internal ME
    CSV_HW_GROUP_INTERNAL_DISTO        = 23, // CS - Internal Disto
    CSV_HW_GROUP_IMU                   = 24, // GS - IMU
    CSV_HW_GROUP_TPS_SHIFTTRIBRACH     = 32, // TPS - Shiftable Tribrach
    CSV_HW_GROUP_TPS_AUTOHEIGHT        = 33, // TPS - AutoHeight
    CSV_HW_GROUP_TPS_DYNAMICLOCK       = 34, // TPS - DynamicLock
};

// Licences information
enum CSV_PROPERTY {

    CSV_PROPERTY_DEMO_LICENSE          = 23, // Demo Licence
    CSV_PROPERTY_GEOCOM_ROBOTICS       = 25, // GeoCom Robotics
    CSV_PROPERTY_GEOCOM_IMAGING        = 26, // GeoCom Imaging
    CSV_PROPERTY_GEOCOM_SCANNING       = 34, // GeoCom Scanning
    CSV_PROPERTY_GEOCOM_VIDEO          = 35, // GeoCom Video
    CSV_PROPERTY_VIRTUAL_GEOCOM        = 37, // Virtual GeoCom
    CSV_PROPERTY_AUTHENTICATION_ALLOWED = 41, // GeoCom Authentication
    CSV_PROPERTY_DYNAMIC_LOCK          = 44, // DynamicLock
    CSV_PROPERTY_SEARCH_AND_LOCK       = 45, // Search and Lock
};
```

```
// Loc8 status information
enum CSV_LOC8_STATUS_MASK {
    LOC8_STATUS_OK = 0x00, // OK
    LOC8_DEVICE_VALIDATION_FAILED = 0x01, // Failed to validate LOC8
                                         // device Id
    LOC8_COMMUNICATION_ERROR = 0x02, // Error in communication to
                                         // LOC8 device
    LOC8_LOCKED = 0x04, // LOC8 device has status
                        // LOCKED and will shut down
    LOC8_ALARM_ACTIVATED = 0x08, // LOC8 device alarm is
                        // activated
    LOC8_DEVICE_NOT_RESPONDING = 0x10, // LOC8 device does not
                                         // respond. Total station will
                                         // shut down
    LOC8_DEVICE_NOT_ATTACHED = 0x20 // No LOC8 device attached to
                                         // Total station. Instrument
                                         // will shut down
}
```

6.1.2 BAP

```
// Distance measurement programs
enum BAP_USER_MEASPRG {
    BAP_CONT_REF_STANDARD // Points to BAP_CONT_REF_SYNCHRO by default
};
```

6.1.3 SCN

```
// New Scan rates are available
enum SCN_SCAN_RATE_TYPE {
    SCN_RATE_FULL = 0,
    SCN_RATE_MIDDLE = 1,
    SCN_RATE_LOW = 2,
    SCN_RATE_ADAPTIVE = 3,
    SCN_RATE_2KHZ = 4,
    SCN_RATE_4KHZ = 5,
    SCN_RATE_8KHZ = 6,
    SCN_RATE_16KHZ = 7,
    SCN_RATE_30KHZ = 8
};

// TPS scan axis type
enum SCN_SCAN_AXIS_TYPE {
    SCN_SCAN_AXIS_HZ = 0 // Horizontal - only supported on new MS60
};
```

6.1.4 AUT

```
// Define what to progress after loss of lock has issued
enum AUT_SEARCH_KIND {
    SK_NoSearch    = 0, // Wait & Lock
    SK_ATR         = 1, // Start fine search with ATR
    SK_PS          = 2, // Start CubeSearch
    SK_LastPoint   = 3, // Turn to last measured point
    SK_StopSearch  = 4  // Stop search
};
```

6.2 CHANGED FUNCTIONS

6.2.1 CSV_GetHardwareOption – Get hardware option of the device

ASCII-Request

%R1Q, 5140:eHwGroup

ASCII-Response

%R1P, 0, 0:RC,rlValue

Remarks

Parameters

eHwGroup	In	CSV_HW_GROUP - Hardware Key Id
rlValue	Out	Value of the key indicates if the key is available. Some are binary, some have different types of hardware

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.2 CSV_CheckProperty – Check licence property

ASCII-Request

%R1Q, 5139:eProperty

ASCII-Response

%R1P, 0, 0:RC,bAvailable

Remarks

Parameters

eProperty	In	CSV_PROPERTY - Check if a Licence is loaded and active on the device
bAvailable	Out	Indication if the Licence is available or not

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.3 CSV_GetLOC8BatteryPercentage – Check battery state

ASCII-Request

%R1Q, 5218:

ASCII-Response

%R1P, 0, 0:RC,batteryStatus

Remarks

The battery status is only provided once on start of the instrument. To request a new state, the instrument needs to be restarted

Parameters

batteryStatus	Out	String - Battery status of LOC8 device in percentage
---------------	-----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.4 CSV_GetLOC8Id – Get the device id

ASCII-Request

%R1Q, 5216:

ASCII-Response

%R1P, 0, 0:RC,deviceId

Remarks

Parameters

deviceId	Out	String - unique id of the LOC8 device
----------	-----	---------------------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.5 CSV_GetLOC8Imei – Get the device IMEI

ASCII-Request

%R1Q, 5217:

ASCII-Response

%R1P, 0, 0:RC,imei

Remarks

This is the IMEI of the GSM module of the LOC8 device.

Parameters

imei	Out	String - IMEI of GSM module (LOC8)
------	-----	------------------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.6 CSV_GetLOC8Status – Get status of LOC8 device

ASCII-Request

%R1Q, 5236:

ASCII-Response

%R1P, 0, 0:RC,status

Remarks

The battery status is only provided once on start of the instrument. To request a new state, the instrument needs to be restarted

Parameters

status	Out	CSV_LOC8_STATUS_MASK - current status of LOC8 device
--------	-----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.7 SCN_SetScanFastAxis – Set the rotating axis definition for TPS scan

ASCII-Request

%R1Q, 23834:eFastAxis

ASCII-Response

%R1P, 0, 0:RC

Remarks

The rotating axis for the defined scan. **New MS60 instruments require to define SCN_SCAN_AXIS_HZ** before start scanning.

Parameters

eFastAxis	In	SCN_SCAN_AXIS_TYPE - Axis type to be used for scan
-----------	----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.8 SCN_CreateScan – Create a new scan**ASCII-Request**

%R1Q,23808:eScanType,eDeviceType

ASCII-Response

%R1P,0,0:RC

Remarks

Parameter 1 supports now partial scans within type SCN_DOME. This will create a scan with an upper & lower boundary defined by SCN_AddScanPoint to only scan a partial area. If SCN_DOME is used on a new MS60, setting SCN_SetScanFastAxis to SCN_SCAN_AXIS_HZ is mandatory.

Parameters

eScanType	In	SCN_SCAN_DEF_TYPE - Type of scan to be executed
eDeviceType	In	SCN_MEMORY_TYPE - Memory type where to store the scan

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

Example

// Create scan

%R1Q,23808:1,2

// Add points to define upper and lower boundary for band scan

%R1Q,23810:0,1.256637061

%R1P,0,0:0

%R1Q,23810:0,1.884955592

%R1P,0,0:0

// Set resolution

%R1Q,23811:0.001570796326795,0.001570796326795

// Set scan rate

%R1Q,23819:0

// Set rotating axis to horizontal

%R1Q,23834:0

// Start scan

%R1Q,23809:

6.2.9 SCN_GetEstimatedScanPoints – Get count of points estimated by this scan

ASCII-Request

%R1Q,23836:

ASCII-Response

%R1P,0,0:RC,uiNumPoints

Remarks

This command returns the points estimated by the system for the defined scan

Parameters

uiNumPoints	Out	Number of estimated points
-------------	-----	----------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.10 SCN_GetEstimatedScanTime – Get time estimated for this scan

ASCII-Request

%R1Q,23835:

ASCII-Response

%R1P,0,0:RC,dTimeInSeconds

Remarks

Returns the estimated time in seconds for the defined scan

Parameters

dTimeInSeconds	Out	Estimated time in seconds
----------------	-----	---------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.11 ATH_MeasureStationHeight – Autoheight measurement

ASCII-Request

%R1Q,30451:

ASCII-Response

%R1P,0,0:RC,dInstrumentHeight

Remarks

Automatically measure the station height. Only supported on instruments with an Autoheight functionality (MS60/TS60/TS16/TS07/TS10)

Before measuring the station height, check if the autoheight is available with CSV_GetHardwareOption

// Check if functionality is available

%R1Q,5140:33

%R1P,0,0:0,33 // Check ok, new MS60 is used

Parameters

dInstrumentHeight	Out	Automatically measured station height
-------------------	-----	---------------------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.12 AUT_FineAdjust6 – Fine adjust with target recognition

ASCII-Request

%R1Q, 9115:dSrchHz,dSrchV,eVisibility

ASCII-Response

%R1P, 0, 0:RC

Remarks

ATA_VISIBILITY enables the “Retry” mode on Captivate devices for better target recognition

Parameters

dSrchHz	In	Search range Hz-axis [rad]
dSrchV	In	Search range V-axis [rad]
eVisibility	In	ATA_VISIBILITY - Enables the "retry" mode for this single command

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.13 AUT_StartFineAdjust2 – Fine adjust with target recognition

ASCII-Request

%R1Q, 9116:dSrchHz,dSrchV,eVisibility

ASCII-Response

%R1P, 0, 0:RC

Remarks

ATA_VISIBILITY enables the “Retry” mode on Captivate devices for better target recognition

Parameters

dSrchHz	In	Search range Hz-axis [rad]
dSrchV	In	Search range V-axis [rad]
eVisibility	In	ATA_VISIBILITY - Enables the "retry" mode for this single command

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.14 AUT_PS_SearchWindowAndLock – Search target and lock

ASCII-Request

%R1Q, 9118:

ASCII-Response

%R1P, 0, 0:RC

Remarks

This command starts PowerSearch inside the given PowerSearch window (defined by AUT_SetSearchArea and optional by AUT_PS_SetRange) and performs a lock to the next target found.

Parameters

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.15 AUT_PS_SearchNextAndLock – Search next target and lock in

ASCII-Request

%R1Q,9105:lDirection,bSwing

ASCII-Response

%R1P,0,0:RC

Remarks

This command executes the 360° default PowerSearch and searches for the next target and performs a lock to the next target found. A previously defined PowerSearch window AUT_SetSearchArea is not taken into account. Use AUT_PS_SearchWindow to do so

Parameters

lDirection	In	Defines the searching direction (CLKW=1 or ACLKW=-1)
bSwing	In	TRUE: Searching starts -10 gon to the given direction Direction. This setting finds targets left of the telescope direction faster

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.16 AUT_GetLossOfLockSearchKind – Search Kind

ASCII-Request

%R1Q,9068:

ASCII-Response

%R1P,0,0:RC,eSearchKind

Remarks

Gets the kind of search to be executed after a Loss of Lock.

Parameters

eSearchKind	Out	AUT_SEARCH_KIND - type of search to be executed after loss of lock
-------------	-----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.17 AUT_SetLossOfLockSearchKind – Search Kind

ASCII-Request

%R1Q,9067:eSearchKind

ASCII-Response

%R1P,0,0:RC

Remarks

Sets the kind of search to be executed after a Loss of Lock.

Parameters

eSearchKind	In	AUT_SEARCH_KIND - type of search to be executed after loss of lock
-------------	----	--

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.18 AUT_PS_SearchLostLock – Cube search

ASCII-Request

%R1Q,9080:unSearchCounter

ASCII-Response

%R1P,0,0:RC

Remarks

This command executes a stand alone Cube Search, the same like after a loss of lock. The movement direction is taken into account. *unSearchCounter* defines the count of retries to be done (not implemented, default is 1).

Parameters

unSearchCounter	In	Number of searches to be done
-----------------	----	-------------------------------

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.19 AUT_ScanSearchVertical – Execute vertical search**ASCII-Request**

%R1Q,9119:hzAngle,backPosit,vAngle_Limit1,vAngle_Limit2

ASCII-Response

%R1P,0,0:RC,numberOfTargetsFound

Remarks

This command performs a vertical search of the telescope either in full movement or in the defined limits within vAngle_Limit1/2. The search within the defined limits starts always from bottom to top. BackPosit will move back the telescope after search to the start position (hzAngle).

The field of view of the ATR needs to be considered especially for subsequent vertical searches:

- TS16/TS60/MS60 FoV: 1°25' (0.0247255 rad)
- TM60 FoV: 0°28' (0.0081449 rad)

It can be used with every Captivate based Totalstation with ATR – even without powersearch (TM60) at least firmware version 6.00.

Parameters

hzAngle	In	The Hz-Angle where the vertical search will be executed [rad]
backPosit	In	TRUE(1): posit back the telescope where the command was send to start the vertical search (hzAngle) FALSE(0): telescope stays at the end of the vertical search, no posit.
vAngle_Limit1/vAngle_Limit2	In	[Optional parameter]: If not used, the full vertical movement of the telescope will be performed. Otherwise the lower value will be used as upper limit and the higher value as lower limit
numberOfTargetsFound	Out	Number of targets found in vertical search

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.20 AUT_ScanSearchVerticalGetTarget – Execute vertical search**ASCII-Request**

%R1Q,9120:targetNumber

ASCII-Response

%R1P,0,0:RC,numberOfTargetsFound,targetOutput

Remarks

This command returns the Hz and V-Angle of the target measured by the ATR. The target is measured “on-the-fly” what is resulting in less accuracy of the angle pair because the movement of the ATR is not stopped during

measurement. For higher accuracy it is recommend to exactly aim at the target with AUT_FineAdjust6, especially for close ranges.

It can be used with every Captivate based Totalstation with ATR – even without powersearch (TM60) at least firmware version 6.00.

Parameters

targetNumber	In	The target to be found
numberOfTargets	Out	The number of the targets found
targetOutput	Out	OUT_TARGET_TYPE: targetOutput: HzAngle,VAngle,SlopeDist %R1P,0,0:RC,1,1.51100,0.92221,12.882

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

Example (ASCII)

```
// Search vertical for targets
%R1Q,9119:2.211824598077613,1,2,1
%R1P,0,1:0,2
```

```
// Get Angles from target 1
%R1Q,9120:0
%R1P,0,2:0,2,2.214276334643328,1.783728315537273,119.223
```

```
// Get Angles from target 2
%R1Q,9120:1
%R1P,0,3:0,2,2.211961730356193,1.6485565337829,122.401
```

6.2.21 BAP_SearchTarget2 - With enhanced target recognition for misty rain conditions

ASCII-Request

```
%R1Q,17042:eVisibility
```

ASCII-Response

```
%R1P,0,0:RC
```

Remarks

ATA_VISIBILITY enables the “Retry” mode on Captivate devices for enhanced target recognition

Parameters

eVisibility	In	ATA_VISIBILITY - Enables the "retry" mode for this single command
-------------	----	---

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.22 BAP_MeasDistanceAngle2 – Measurement with enhanced target recognition for misty rain conditions

ASCII-Request

```
%R1Q,17041:eVisibility
```

ASCII-Response

```
%R1P,0,0:RC,dHz,dV,dDist,dDistMode
```

Remarks

Enables the “Retry” mode on Captivate devices for enhanced target recognition for this single command

Parameters

eVisibility	In	ATA_VBISIBILITY - Enables the "retry" mode for this single command
dHz	Out	Hz Angle of the target
dV	Out	VAngle of the target
dDist	Out	Slope Distance to the target
dDistMode	Out	BAP_USER_MEASPRG - Measurement program

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.23 TMC_DoMeasure3 – Measurement with enhanced target recognition for misty rain conditions**ASCII-Request**

%R1Q,2179:eCommand,eMode,eVisibility

ASCII-Response

%R1P,0,0:RC

Remarks**Parameters**

eCommand	In	BAP_USER_MEASPRG - Measurement program
eMode	In	TMC_INCLINE_PRG - Inclination program
eVisibility	In	Enables the "retry" mode for this single command

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.24 TMC_DoMeasure4 – Measurement with enhanced target recognition for misty rain conditions**ASCII-Request**

%R1Q,2180:eCommand,eVisibility

ASCII-Response

%R1P,0,0:RC

Remarks**Parameters**

eCommand	In	BAP_USER_MEASPRG - Measurement program
eVisibility	In	ATA_VISIBILITY - Enables the "retry" mode for this single command

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

6.2.25 TMC_DoMeasure5 – Measurement with enhanced target recognition for misty rain conditions**ASCII-Request**

%R1Q,2182:eCommand,eMode,eVisibility,eSessionBehavior

ASCII-Response

%R1P,0,0:RC

Remarks**Parameters**

eCommand	In	BAP_USER_MEASPRG - Measurement program
eMode	In	TMC_INCLINE_PRG - Inclination program
eVisibility	In	ATA_VISIBILITY - Enables the "retry" mode for this single command
eSessionBehavior	In	Stop session on GeoCom idle

Return-Code Names and Return-Code Values

GRC_OK	0	Execution successful.
--------	---	-----------------------

7 APPENDIX – RETURN CODES

7.1 ATH

RetCodeName	Value	Hexadecimal	Description
GRC_ATH_SYSTEM_ERR	16897	0x4201	Fatal ATH sensor error
GRC_ATH_COMMUNICATION_ERROR	16898	0x4202	General communication error
GRC_ATH_TEMPERATURE_TOO_HIGH	16899	0x4203	Temperature too high
GRC_ATH_TEMPERATURE_TOO_LOW	16900	0x4204	Temperature too low
GRC_SIGNAL_RECEIVED_TOO_LOW	16901	0x4205	Signal received from target too low
GRC_SIGNAL_RECEIVED_TOO_HIGH	16902	0x4206	Signal received from target too high
GRC_TARGET_TO_NEAR_OR_TO_FAR_AWAY	16903	0x4207	Target is too near or too far away
GRC_OPTIC_DISTURBANCE_IN_LASER	16904	0x4208	Optical disturbance in laser beam